

Density + Rearranging Equations

Test · 24 marks total

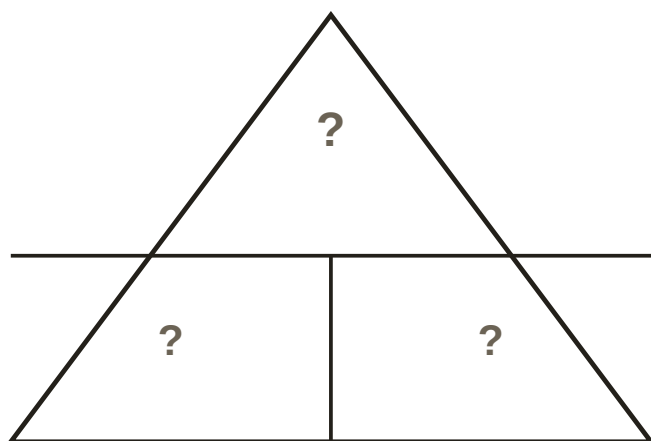
Name: _____

Date: _____

Score: _____ / 24

Section A — Complete the formula triangle

Fill in the three blanks with ρ , m and V in the correct places.



[3 marks]

Section B — Straight substitution

B1. $m = 30 \text{ kg}$, $V = 6 \text{ m}^3$. Calculate the density.

[2 marks]

B2. $m = 45 \text{ g}$, $V = 9 \text{ cm}^3$. Calculate the density.

[2 marks]

Section C — Rearrange and solve

C1. A block has a density of 4 g/cm^3 and a volume of 12 cm^3 . Calculate its mass. Show how you rearranged the equation.

[3 marks]

C2. A liquid has a density of 1.5 g/cm^3 and a mass of 60 g . Calculate its volume. Show how you rearranged the equation.

[3 marks]

C3. A metal has a density of 9 g/cm^3 and a mass of 45 g . Calculate its volume.

[3 marks]

C4. A gas has a density of 0.002 g/cm^3 and occupies 2000 cm^3 . Calculate its mass.

[3 marks]

Section D — Reasoning

D1. Two blocks have the same mass, but block A has a larger volume than block B. Which block has the higher density? Explain your answer.

[2 marks]